

Lucas Garcia

Embedded Systems & IoT Firmware Engineer • 12 Years Experience • MERIQ Index: 78% (Strong Candidate)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: I2C/SPI/UART, ARM Cortex, Python, MQTT, Embedded Linux, FreeRTOS, PCB Debugging

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Qualcomm* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core I2C/SPI/UART service, reducing latency by 48%.
- Led a squad of 3 engineers delivering scalable microservices processing 14M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 68%.

Associate Embedded Systems & IoT Firmware Engineer — *Tesla* (2014 - 2023)

- Designed and maintained ARM Cortex data endpoints with 95% SLA uptime.
- Optimized database indexing and caching layers using Python saving \$34k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [I2C/SPI/UART, ARM Cortex, Python, MQTT]

Engineered end-to-end production solution leveraging I2C/SPI/UART and ARM Cortex with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [Python, MQTT, Embedded Linux, FreeRTOS]

Architected high-resilience system integrating Python with 99.9% uptime.

EDUCATION & CREDENTIALS

M.S. in Software Engineering

Delhi Technological University (DTU) — 2010 - 2014 (Score: 3.7 / 4.0)